

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



A COMPREHENSIVE COVERAGE

MODULE – 43

NANO TECHNOLOGY - 1

NANOTECHNOLOGY

DEFINITION

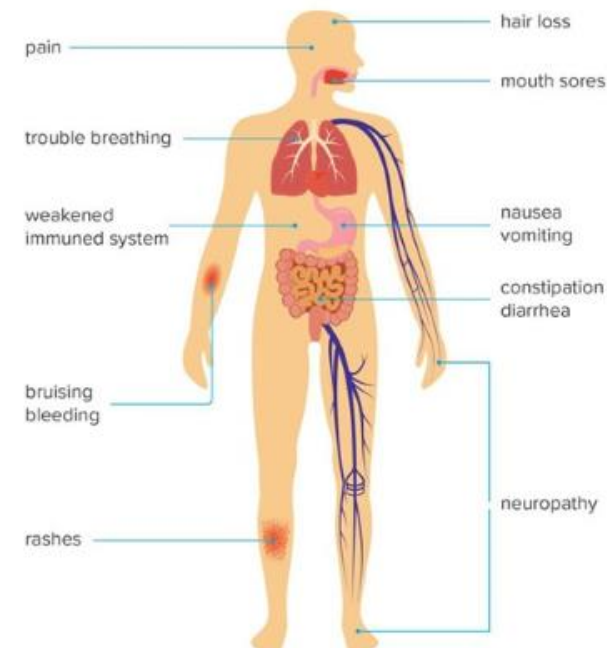
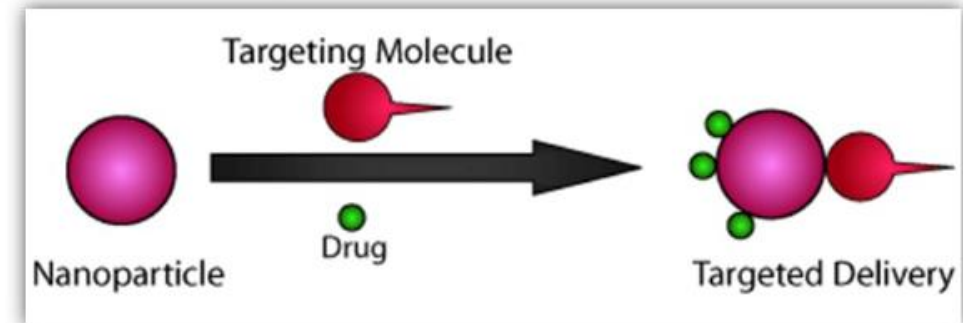
- Nanotechnology is defined as the study and use of structures between **1 nanometer and 100 nanometers in size**. To give you an idea of how small that is, it would take eight hundred 100 nanometer particles side by side to match the width of a human hair.
- The ability to see nano-sized materials has opened up a world of possibilities in a variety of industries and scientific endeavors.
- Because **nanotechnology is essentially a set of techniques that allow manipulation of properties at a very small scale**, it can have many applications



NANOTECHNOLOGY

TARGETED DRUG DELIVERY

- Most harmful side effects of treatments such as chemotherapy are a result of drug delivery methods that don't pinpoint their intended target cells accurately.
- Researchers have been able to attach RNA strands, measuring about 10 nm in diameter, to nanoparticles and **fill the nanoparticles with a chemotherapy drug**. These RNA strands are attracted to cancer cells. **When the nanoparticle encounters a cancer cell it adheres to it and releases the drug into the cancer cell.**
- This directed method of drug delivery has great potential for treating cancer patients while producing **less side harmful affects than those produced by conventional chemotherapy.**



NANOTECHNOLOGY

FABRICS

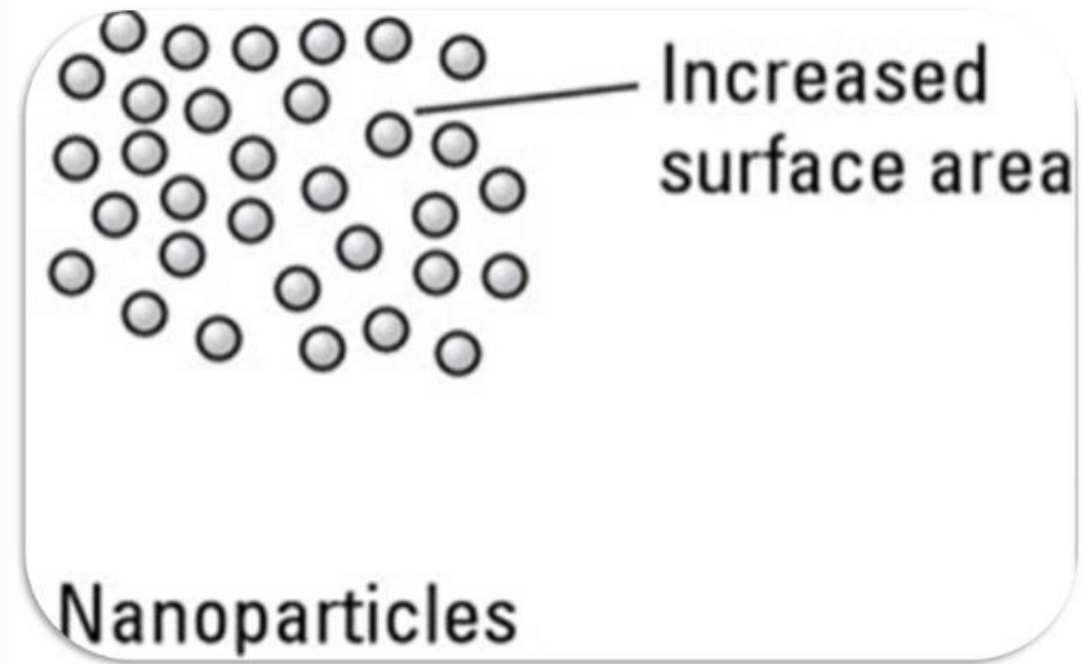
- The properties of familiar materials are being changed by manufacturers who are **adding nano-sized components to conventional materials to improve performance.**
- For example, some clothing **manufacturers are making water and stain repellent clothing** using nanoparticles in the fabric that cause water to bead up on the surface.



NANOTECHNOLOGY

REACTIVITY OF MATERIALS

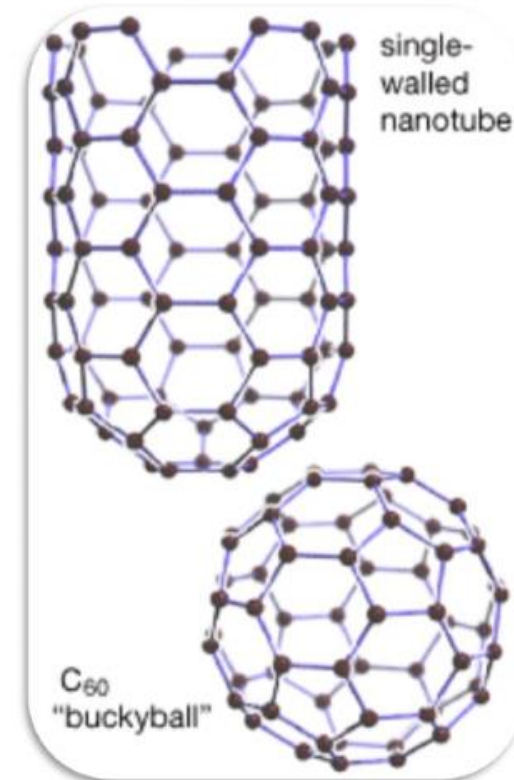
- The properties of many conventional materials change when formed as nano-sized particles.
- This is because **nanoparticles have a greater surface area per weight** than larger particles; they are therefore **more reactive**.
- For example, studies have shown that **nanoparticles of iron can be effective in the cleanup of chemicals in groundwater** because they react more efficiently to those chemicals than larger iron particles.



NANOTECHNOLOGY

STRENGTH OF MATERIALS

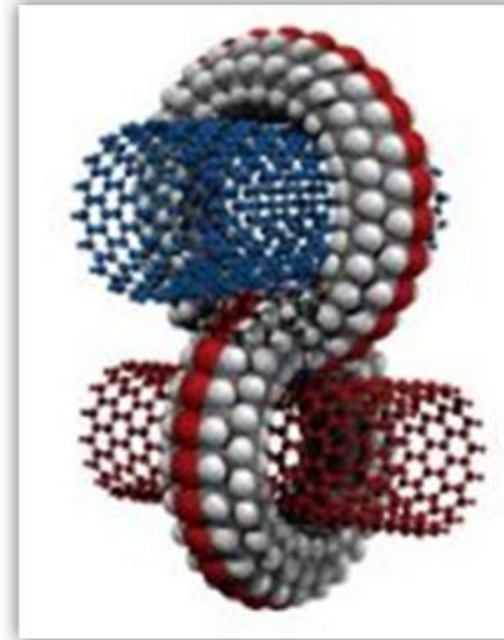
- Nano-sized particles of carbon, (nanotubes and bucky balls) are **extremely strong**.
- Nanotubes and bucky balls are composed of only carbon and their **strength comes from special characteristics of the bonds between carbon atoms**.
- One proposed application that shows the strength of nanoparticles of carbon is the **manufacture of t-shirt weight bullet proof vests made out of carbon nanotubes**.



NANOTECHNOLOGY

MOLECULAR MANUFACTURING

- A molecular fabricator is a device that uses tiny manipulators to position atoms and molecules to build an object as complex as a desktop computer.
- Researchers believe that raw materials can be used to reproduce almost any inanimate object using this method.

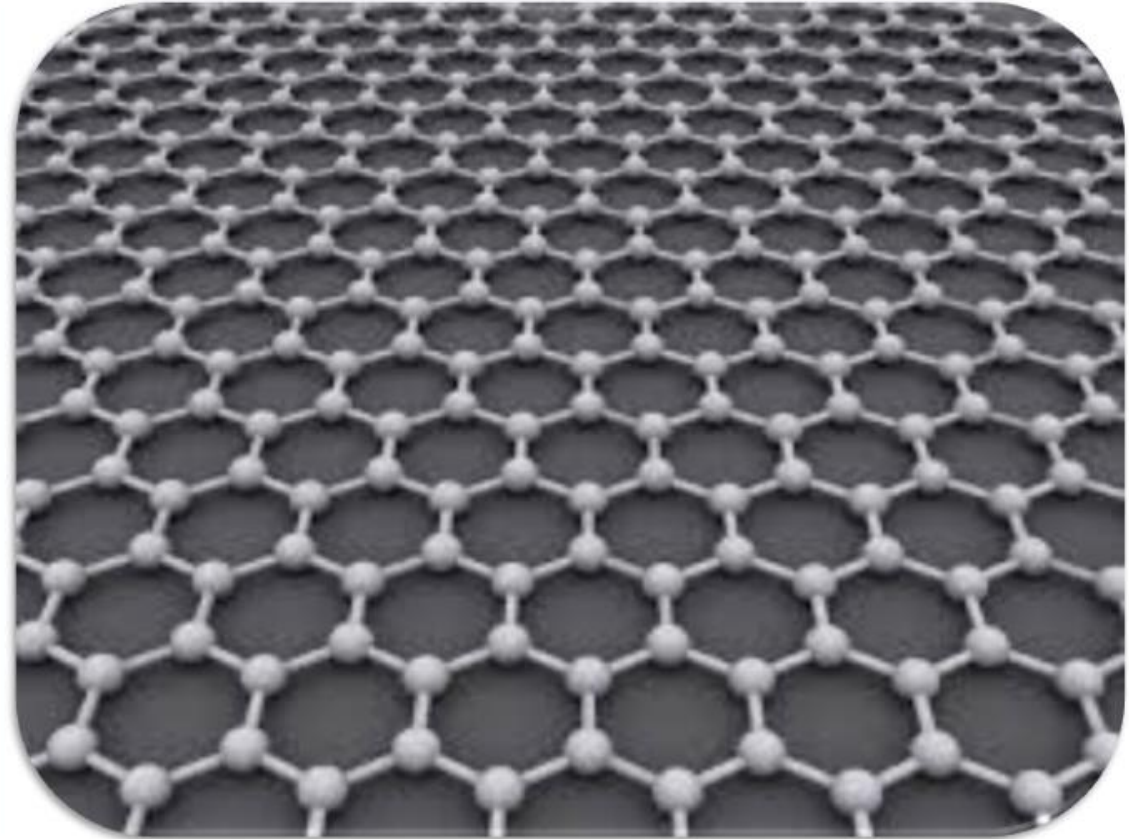


NANOTECHNOLOGY

NANOMATERIALS

1 GRAPHENE

- Graphene is a sheet of hexagonally bonded carbon atoms. It is the thinnest material known to man at **one atom thick**
- Scientists say that it can be **used to create materials 200 times stronger than steel**, materials better at conducting both heat and electricity than copper and sufficiently flexible materials which can be used to form **super-strength "tubes" for use in construction.**



POTENTIAL APPLICATIONS

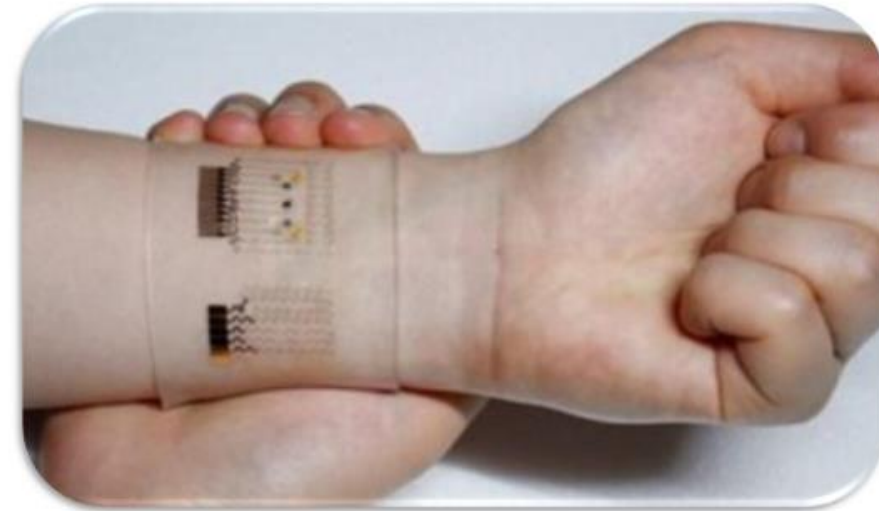
LOWER COST OF DISPLAY SCREENS IN MOBILE DEVICES

Researchers have found that graphene can be used in Organic Light Emitting Diodes (OLED). The use of graphene reduces the cost, eliminates the use of metals in the OLED, which may make devices easier to recycle.



GLUCOSE MEASUREMENT

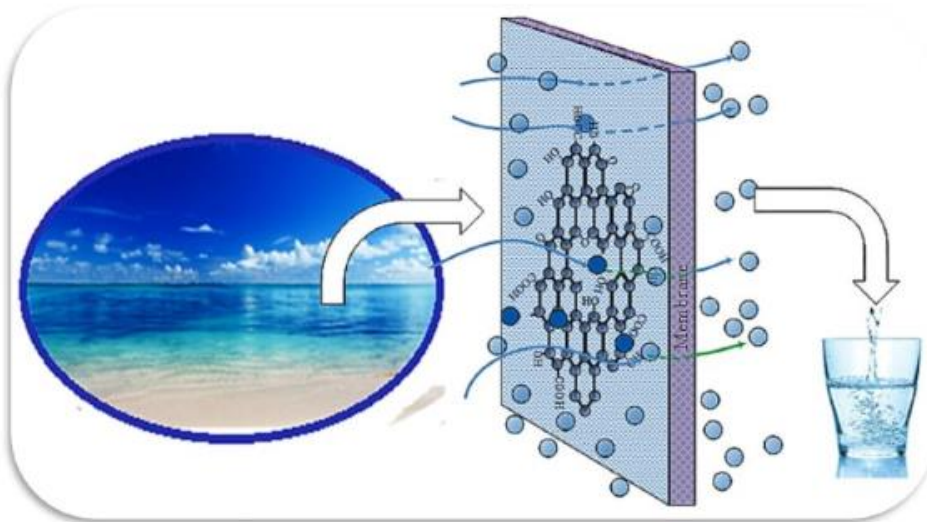
Researchers are developing a graphene based sensor to measure glucose levels without requiring a finger prick blood test.



POTENTIAL APPLICATIONS

LOW COST WATER DESALINATION

Researchers have determined that **graphene can be used to remove ions from water**. They believe this can be used to desalinate sea water **at a lower cost** than the reverse osmosis techniques currently in use.



LIGHTWEIGHT NATURAL GAS TANKS

Researchers have developed a **composite material using plastic and graphene that blocks the passage of gas molecules**. This material may be used in applications ranging from **soft drink bottles to lightweight natural gas tanks**.



POTENTIAL APPLICATIONS

SENSORS TO DIAGNOSE DISEASES

These sensors are based upon graphene's large surface area and the fact that molecules that are sensitive to particular diseases can attach to the carbon atoms in graphene.

ULTRACAPACITORS WITH BETTER PERFORMANCE THAN BATTERIES

These ultra capacitors store electrons on graphene sheets, taking advantage of the large surface of graphene to provide an increase in the electrical power that can be stored in the capacitor.

Researchers are projecting that these ultra capacitors will have as much electrical storage capacity as lithium-ion batteries but will be able to be recharged in minutes instead of hours.



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